

PHAN NGUYEN HUONG LE

📞 715-440-1726 ✉ hpl14@scarletmail.rutgers.edu www.linkedin.com/in/hpl14 github.com/valle1306

Research Interests

uncertainty quantification, probabilistic modeling, and computational approaches to learning, decision-making, and cognition.

Education

Rutgers University

M.S. in Statistics- Data Science

Jan 2024 – May 2026

New Brunswick, NJ

Rutgers University

B.A.Sc. in Biomathematics

Jan 2020 – May 2024

New Brunswick, NJ

Research Experience

R/ Shiny Developer - Rutgers University — June 2025- Present

New Brunswick, NJ

- Built an interactive Shiny application under Dr. Han Zhang and Dr. Philip He's guidance to visualize hypothesis graphs and redistribute test weights dynamically for multiple testing procedures.
- Collaborated on extending the open-source trialsimulator R package, improving usability and reproducibility for clinical trial simulations.
- Designed features for node/edge editing, JSON import/export, enabling flexible experimental design.
- Engaged in iterative development with faculty feedback, focusing on accessibility for applied statisticians and biomedical researchers.

Directed Reading in Causal Inference — June 2025- August 2025

New Brunswick, NJ

- Studied CBPS, AIPW/DR estimators, overlap/trim rules, and matching vs weighting methods, mentored by Dr. Nicole Pashley.

Imaging Data Analyst - University of Pennsylvania/ CHOP — July 2024- Present

Philadelphia, PA

- Developed and deployed nnU-Net v2 pipelines on HPC clusters (Slurm/GPU) for pediatric brain tumor segmentation.
- Implemented Monte Carlo Dropout with 20 inference passes per case to generate uncertainty maps, supporting clinician quality assurance.
- Processed and curated 100+ GB of multi-sequence MRI data with automated ETL, artifact logging, and reproducibility checks.
- Contributed to AutoSurv pipeline integrating radiomics (1.2k+ features/patient) and clinical data; applied KL-PMVAE + LFSurv models and DeepSHAP interpretability, achieving 10% higher concordance than Cox baselines.

Research Assistant — Social Perception Lab — Sept 2023- May 2024

New Brunswick, NJ

- Performed data cleaning and transformation for psychometric scale evaluation studies (Likert-type responses).
- Applied Confirmatory Factor Analysis (CFA) in R to assess construct validity and model fit.
- Contributed to literature review and synthesis on measurement reliability and bias in social psychology, mentored by Dr. Lee Jussim.
- Collaborated with senior researchers to interpret CFA results and refine scale design for ongoing projects.

Research Assistant — ABUSA REHAB Health Psychology Lab — Jan 2022- May 2023 *New Brunswick, NJ*

- Conducted participant phone screens and managed data collection for smoking and stress intervention studies led by Dr. Teresa Leyro and Dr. Samantha Farris.
- Processed heart rate data using HRV and PRV software to extract physiological stress biomarkers.
- Supported clinical research protocols by maintaining participant logs, preparing data for statistical analysis, and assisting in literature review.
- Gained exposure to translational research linking behavioral interventions with psychophysiological outcomes.

Work Experience

Bioinformatics Data Scientist- Bristol Meyers Squibb Challenge

New Brunswick, NJ

- Designed an interactive Streamlit-based computation tool to estimate charge variants in therapeutic proteins, integrating pI calculation and PTM site modeling to assess formulation stability.
- Optimized a Python workflow for automated charge estimation, visualization, and reporting to enhance biopharmaceutical quality analysis and formulation design.

Data Science Intern - IDX Exchange

New York, NY

- Ingested and cleaned 2.3M MLS rows from 8 data sources into analytics tables; automated daily refresh with 99.5% success.
- Trained an XGBoost AVM for home values with geo features (schools/transit/parks, neighborhood indices, age/reno flags), cutting RMSE 18% and reaching MAPE 7.9% on 5-fold CV (R^2 0.71 $\rightarrow$ 0.77).
- Deployed a home-value estimator API and integrated outputs into the property search module.
- Built Tableau market dashboards (price/sqft, DOM, absorption) across 42 counties; adopted by 3 product teams, reducing ad-hoc reporting time 60%.

Quantitative Research Mentee - JPMorgan Chase & Co.

New York, NY

- Engineered market-data ETL in Python (`yfinance`) to date-partitioned Parquet with schema/type checks and daily ingestions; ran 1M+ Monte Carlo stress paths, reducing VaR error by 8%.
- Built forecasting/pricing components: LSTM improved equity prediction accuracy 12%; vectorized Black-Scholes pricer increased throughput 15%.

Presentations

- “Uncertainty Quantification in Imaging Segmentation” — Presented lab reading on recent methods in Monte Carlo Dropout and predictive entropy, Biomedical Engineering Lab, Univ. of Pennsylvania/CHOP (2025).
- “Interactive R Shiny App for Graphical Multiple Testing Procedures” — Demoed app features and statistical design functionality for the TrialSimulator package, Department of Statistics Mini-Symposium, Rutgers University (2025).

Teaching and Mentoring

Learning Assistant, Calculus II — Rutgers University

Fall 2023, Spring 2024

- Led weekly problem-solving sessions for 40 students, reinforcing multivariable calculus concepts.
- Provided one-on-one tutoring, improving student comprehension and exam performance.
- Collaborated with course instructors to align review materials with lecture content.

Grader, Undergraduate Statistics Courses - Rutgers University

Fall 2024, Spring 2025, Fall 2025

- Evaluated assignments and exams for 120 students, ensuring consistency and fairness in assessment.
- Supported instructional team by maintaining organized grade records and providing structured feedback.

Extracurricular Activities

- Orientation Leader at Rutgers University, fostering leadership and communication through guiding over 2000 students.
- Active Member of Data Science and Quantitative Finance Clubs, participating in data-driven & financial projects.
- Lead Translator — Goc Tam Ly Hoc (Psychology Education Page, Vietnam): Directed a year-long translation initiative, producing accessible psychology education content for a broad public audience.

Relevant Courses

Master’s Courses: Regression and Time Series, Probability and Inferences, Statistical Learning, Statistical Modeling, Data Mining, Data Wrangling, Database Management, Risk Practice, Advanced Statistical Methods in Finance

Undergraduate Courses: Intro to CS, Data Structures, Math Theory of Probability, Math Theory of Statistics, Linear Algebra, Applied Multivariate, Bayesian Data Analysis, Combinatorics Theory, Calculus II

Technical Skills

Programming Languages: Python, R, SQL